# Chapter 6

# Parsing Native Exports

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## 6.1 Learning objectives

After completing this chapter, you will be able to:

- Parse BibTeX files using `bib2df` and `RefManageR`
- Import Web of Science plain-text and Scopus CSV exports via `bibliometrix`
- Standardize field names across formats into a unified schema
- Diagnose common parsing failures and missing fields

## 6.2 Setup

```
library(tidyverse)
library(bib2df)
library(RefManageR)
library(bibliometrix)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

## 6.3 Conceptual background

Not all bibliometric data comes from an API. Researchers frequently begin with files exported from database search interfaces: BibTeX (`.bib`) from Google Scholar or Zotero, RIS (`.ris`) from Scopus or PubMed, plain text from Web of Science, or CSV from Scopus. Each format encodes the same information — authors, title, year, journal, DOI — in different structures.

The `bibliometrix` package (Aria and Cuccurullo, 2017) provides `convert2df()`, a versatile function that reads multiple export formats and returns a standardized data frame. For BibTeX specifically, `bib2df` and `RefManageR` offer complementary parsing capabilities. The key challenge is not parsing per se but standardization: ensuring that author names, journal titles, and identifiers are represented consistently regardless of the source format.

## 6.4 Worked example

### 6.4.1 Parsing a BibTeX string

We construct a small BibTeX string inline to demonstrate parsing without external files.

```
bib_text <- '
@article{smith2023,
  author = {Smith, John A. and Doe, Jane B.},
  title = {Advances in Bibliometric Methods},
  journal = {Journal of Informetrics},
  year = {2023},
  volume = {17},
  pages = {101--115},
  doi = {10.1234/example.2023}
}
@article{chen2022,
  author = {Chen, Wei and Li, Xiao},
  title = {Citation Networks in Computer Science},
  journal = {Scientometrics},
  year = {2022},
  volume = {127},
  pages = {3201--3220},
  doi = {10.1234/example.2022}
}
'

tmp_bib <- tempfile(fileext = ".bib")
```

```
writeLines(bib_text, tmp_bib)

bib_df <- bib2df(tmp_bib)
glimpse(bib_df)
```

```
#> Rows: 2
#> Columns: 27
#> $ CATEGORY     <chr> "ARTICLE", "ARTICLE"
#> $ BIBTEXKEY    <chr> "smith2023", "chen2022"
#> $ ADDRESS      <chr> NA, NA
#> $ ANNOTE       <chr> NA, NA
#> $ AUTHOR       <list> <"Smith, John A.", "Doe, Jane B.">,
 ↪  <"Chen, Wei", "Li, Xi~
#> $ BOOKTITLE    <chr> NA, NA
#> $ CHAPTER      <chr> NA, NA
#> $ CROSSREF     <chr> NA, NA
#> $ EDITION      <chr> NA, NA
#> $ EDITOR       <list> NA, NA
#> $ HOWPUBLISHED <chr> NA, NA
#> $ INSTITUTION  <chr> NA, NA
#> $ JOURNAL      <chr> "Journal of Informetrics",
 ↪  "Scientometrics"
#> $ KEY          <chr> NA, NA
#> $ MONTH        <chr> NA, NA
#> $ NOTE         <chr> NA, NA
#> $ NUMBER       <chr> NA, NA
#> $ ORGANIZATION <chr> NA, NA
#> $ PAGES        <chr> "101--115", "3201--3220"
#> $ PUBLISHER    <chr> NA, NA
#> $ SCHOOL       <chr> NA, NA
#> $ SERIES       <chr> NA, NA
#> $ TITLE        <chr> "Advances in Bibliometric Methods",
 ↪  "Citation Networks in~
#> $ TYPE         <chr> NA, NA
#> $ VOLUME       <chr> "17", "127"
#> $ YEAR         <dbl> 2023, 2022
#> $ DOI          <chr> "10.1234/example.2023",
 ↪  "10.1234/example.2022"
```

### 6.4.2 Parsing with RefManageR

```
bib_rm <- ReadBib(tmp_bib)
as.data.frame(bib_rm) |> select(title, author, year, journal,
 ↪  doi)
```

```
#>                                          title
 ↪  author
```

```
#> smith2023      Advances in Bibliometric Methods John A. Smith
 ↪  and Jane B. Doe
#> chen2022  Citation Networks in Computer Science          Wei
 ↪  Chen and Xiao Li
#>           year                 journal                  doi
#> smith2023 2023 Journal of Informetrics 10.1234/example.2023
#> chen2022  2022          Scientometrics 10.1234/example.2022
```

### 6.4.3 Parsing WoS plain text with bibliometrix

```
# Requires a WoS export file — not bundled
# wos_df <- convert2df("savedrecs.txt", dbsource = "wos", format
 ↪  = "plaintext")
# glimpse(wos_df)
```

### 6.4.4 Standardizing field names

```
standardize_bib <- function(df) {
  df |>
    transmute(
      title = TITLE,
      authors = AUTHOR,
      year = as.integer(YEAR),
      journal = JOURNAL,
      doi = DOI
    )
}

bib_df |> standardize_bib()
```

```
#> # A tibble: 2 x 5
#>   title                                authors    year journal
 ↪  doi
#>   <chr>                                  <list>    <int> <chr>
 ↪  <chr>
#> 1 Advances in Bibliometric Methods      <chr [2]>  2023
 ↪  Journal of Inform~ 10.1~
#> 2 Citation Networks in Computer Science <chr [2]>  2022
 ↪  Scientometrics     10.1~
```

### 6.4.5 Visualization

```
completeness <- bib_df |>
  summarise(across(everything(), ~ mean(!is.na(.)))) |>
  pivot_longer(everything(), names_to = "field", values_to =
   ↪  "present") |>
  filter(present > 0) |>
  mutate(field = fct_reorder(field, present))

ggplot(completeness, aes(x = present, y = field)) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "Completeness", y = NULL) +
  theme_sci()
```

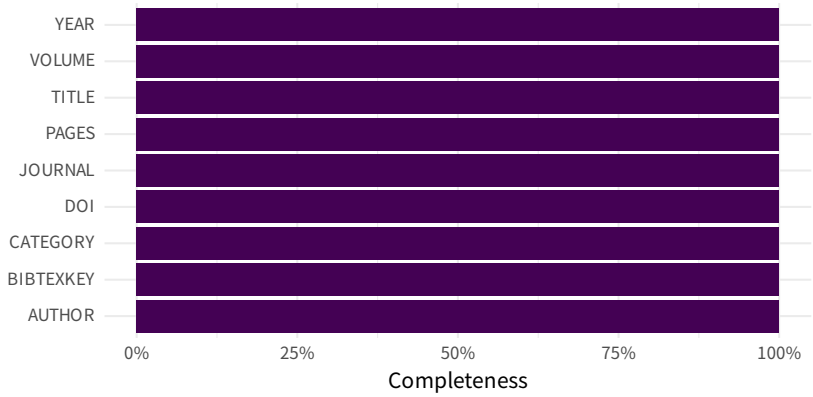



Figure 6.1: Field completeness in the parsed BibTeX records.

## 6.5 Diagnostics and interpretation

After parsing, always check:

- **Row count**: Does the number of records match the export? Missing entries may indicate parsing errors.
- **Field completeness**: Which fields are populated? Abstracts and keywords are often missing in BibTeX exports.
- **Author format**: Names may appear as “Last, First” or “First Last” depending on the source. Standardize before analysis.
- **Encoding**: Non-ASCII characters (diacritics, CJK) can corrupt during parsing. Ensure UTF-8 encoding.

## 6.6 Limitations and responsible use

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- Export files are snapshots: they reflect the database state at export time and become stale immediately.
- Field mapping between formats is imperfect. Some formats lack fields (RIS has no standard abstract tag across all databases).
- Parsing errors are silent — always validate record counts and spot-check metadata against the source (Hicks et al., 2015).

## 6.8 Common pitfalls

## 6.9 Common pitfalls

- **Encoding mismatches**: Exporting as Latin-1 and reading as UTF-8 (or vice versa) corrupts names and titles.
- **Truncated exports**: Many databases cap exports at 500 or 1,000 records per file. Large corpora require multiple export batches.
- **Inconsistent author delimiters**: BibTeX uses "and", RIS uses newlines, CSV uses semicolons. Parsing must handle each.
- **Assuming format from extension**: A `.txt` file may be WoS plain text or something else entirely. Check the first few lines.

## 6.10 Exercises

1. **Export and parse.** Export 50 records from Google Scholar as BibTeX. Parse them with `bib2df`. How many fields are populated?

2. **Format comparison.** Export the same set of records from Scopus as both CSV and RIS. Parse both and compare the fields available.

3. **Standardization function.** Write a function that takes a data frame from `convert2df()` and returns a tibble with columns: `doi`, `title`, `authors`, `year`, `journal`, `cited_by`.

## 6.11 Solutions

Solutions are provided in E.

## 6.12 Further reading

- Aria and Cuccurullo (2017) — The `bibliometrix` package, including `convert2df()` for multi-format parsing.
- Priem et al. (2022) — OpenAlex as an API-first alternative to file-based workflows.

## 6.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:
 ↪  /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenbl
 ↪  asp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C
 ↪  LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8
 ↪  LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8      LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8
 ↪  LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods
 ↪  base
#>
```

```
#> other attached packages:
#>  [1] bibliometrix_5.5.0 RefManageR_1.4.0   bib2df_1.1.2.0
 ↪  rcrossref_1.2.1
#>  [5] gt_1.4.0           tidytext_0.4.3     glue_1.8.1
 ↪  openalexR_3.1.2
#>  [9] lubridate_1.9.5    forcats_1.0.1      stringr_1.6.0
 ↪  dplyr_1.2.1
#> [13] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2
 ↪  tibble_3.3.1
#> [17] ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gridExtra_2.3.1       httr2_1.3.0           readxl_1.6.0
#>  [4] rlang_1.3.0           magrittr_2.0.5         otel_0.2.0
#>  [7] compiler_4.6.1        systemfonts_1.3.2     vctrs_0.7.3
#> [10] httpcode_0.3.0        pkgconfig_2.0.3
 ↪  fastmap_1.2.0
#> [13] labeling_0.4.3        ca_0.72                utf8_1.2.6
#> [16] promises_1.5.0        rmarkdown_2.32         tzdb_0.5.0
#> [19] xfun_0.62             jsonlite_2.0.0
 ↪  SnowballC_0.7.1
#> [22] later_1.4.8           parallel_4.6.1
 ↪  stopwords_2.3
#> [25] R6_2.6.1              stringi_1.8.9
 ↪  RColorBrewer_1.1-3
#> [28] cellranger_1.1.0      Rcpp_1.1.2
 ↪  bookdown_0.48
#> [31] knitr_1.52            triebeard_0.4.1
 ↪  base64enc_0.1-6
#> [34] httpuv_1.6.17         rentrez_1.2.4         Matrix_1.7-5
#> [37] igraph_2.3.4          timechange_0.4.0
 ↪  tidyselect_1.2.1
#> [40] stringdist_0.9.17     rstudioapi_0.19.0
 ↪  pubmedR_1.0.2
#> [43] yaml_2.3.12           viridis_0.6.5         miniUI_0.1.2
#> [46] humaniformat_0.6.0    curl_8.1.0              qpdf_1.4.1
#> [49] lattice_0.22-9        plyr_1.8.9            shiny_1.14.0
#> [52] withr_3.0.3           S7_0.2.2             askpass_1.2.1
#> [55] evaluate_1.0.5         zip_3.0.2              xml2_1.6.0
#> [58] shinycssloaders_1.1.0  pillar_1.11.1
 ↪  janeaustenr_1.0.0
#> [61] DT_0.34.0             plotly_4.12.1
 ↪  generics_0.1.4
#> [64] rprojroot_2.1.1       hms_1.1.4             scales_1.4.0
#> [67] xtable_1.8-8          contentanalysis_1.1.1  tools_4.6.1
#> [70] data.table_1.18.6.1    tokenizers_0.3.0
 ↪  openxlsx_4.2.9
#> [73] pdftools_3.9.1         fs_2.1.0
 ↪  visNetwork_2.1.4
#> [76] XML_3.99-0.25         grid_4.6.1            bibtex_0.5.3
```

```
#> [79] urltools_1.7.3.1       rscopus_0.9.0
 ↪  dimensionsR_0.0.3
#> [82] bibliometrixData_0.3.0 cli_3.6.6
 ↪  viridisLite_0.4.3
#> [85] gtable_0.3.6           digest_0.6.39
 ↪  ggrepel_0.9.8
#> [88] crul_1.6.0            htmlwidgets_1.6.4     farver_2.1.2
#> [91] htmltools_0.5.9        lifecycle_1.0.5        httr_1.4.9
#> [94] here_1.0.2             mime_0.13
```